RiskPulse

Unified CFO & CISO Operational Risk Dashboard

Published: 7/28/2026

Product Capabilities & Specification Document

1. Executive Summary

RiskPulse is a comprehensive, full-stack risk advisory platform built specifically to integrate corporate financial metrics and cybersecurity posture into a singular dashboard. Designed for Managed Service Providers (MSPs), small business consulting groups, and Small/Medium Enterprises (SMEs), it establishes a continuous visual bridge between liquidity metrics and vulnerability audits. In modern operations, IT security failures immediately impact cash flow, while cash runway constraints directly limit the ability of a business to patch systems, enforce policies, or invest in business continuity. RiskPulse solves this friction by calculating a unified business resilience score and delivering automated, prioritized mitigation tasks.

2. Scoring Model & Formulas

The platform calculates real-time risk severity indexes by running rules-based scripts over active transaction ledgers and network vulnerability reports. The core metric is the Overall Business Health Score (0-100), computed as a weighted balance:

Overall Health Score = 100 - (0.5 * Financial Risk Score + 0.5 * Cybersecurity Risk Score)

Financial Risk Score Criteria (Capped at 100)

- **Current Ratio (< 1.5):** Adds +12 risk points; if the ratio falls under 1.0 (indicating immediate insolvency risk), the score adds +25 points.
- **Quick Ratio (< 1.0):** Adds +8 risk points; under 0.8 adds +15 points, reflecting an inability to clear short-term bills without liquidating physical inventories.
- **Debt Ratio (> 50%):** Adds +10 risk points; over 70% adds +20 points, marking a heavily leveraged capital structure.
- **Cash Runway (< 6 mo):** Adds +12 risk points; under 3 months adds +25 points (payroll default hazard).
- **Customer Concentration (> 25%):** Adds +7 risk points; over 40% adds +15 points, exposing the business to extreme client churn risk.
- **Late Receivables (> 10%):** Adds +5 risk points; over 25% adds +10 points, indicating collections delay bottlenecks.

Cybersecurity Risk Score Criteria (Capped at 100)

- **MFA Enforce Status:** Disabled MFA adds +25 risk points (leading entry channel for corporate identity hijack).
- **Password Policies:** Inactive complex password policy adds +10 risk points.
- **Backup Solutions:** Outdated backup logs add +10 risk points; missing backup policies ('none') add +20 risk points.
- **Endpoint Protection:** Disabled endpoint security firewalls add +15 risk points.
- **Phishing Incidents:** Adds +5 risk points per incident (capped at +15 points).

- **Open Vulnerabilities:** Adds +3 risk points per active CVE vulnerability (capped at +15 points).
- **Patching Latency:** Outdated patches add +7 points; critical unpatched system firmware adds +15 points.
- **Staff Training Gap:** Adds risk points dynamically based on training completion rate: 15 * (1 - completion_rate).

3. Detailed Platform Feature Suite

RiskPulse incorporates multi-tenant dashboard views with active background threat calculations, enabling operators to analyze, report, and secure business units efficiently.

- **Multi-Tenant Client Switcher:** Allows operators to swap seamlessly between different client profiles, loading dedicated charts, balance sheets, and documentation indices.
- **Automated Fraud Audit Ledger:** Audits outgoing outlays in real-time. It flags transaction spikes (>4x average outlay size), off-hours operations (outside 7 AM - 8 PM), and duplicate bills submitted within 48 hours.
- **Mini RAG Knowledge Base:** Indexes uploaded files (agreements, insurance covenants, tax records) using text-chunking and pgvector. Users query documents inside a sidebar assistant, which returns answers with source citations.
- **Ad-hoc PDF Report Engine:** Generates full-scale, multi-page client risk summaries on demand, detailing ratios, security statuses, transaction warnings, and advisor recommendations.

4. Small Business & Consulting Firm Value Proposition

For Small and Medium Enterprises (SMEs)

- **Virtual C-Suite Advisory:** Replaces the need for a full-time CFO or CISO by translating complex financial and technical security logs into a single, unified action list.
- **Working Capital Protection:** Prevents operational cash leakage by flagging double-billing and unauthorized off-hours transaction expenses before they are cleared.
- **Document Interrogation:** Provides instant answers to complex liability queries (e.g., insurance exclusions, service SLA clauses) via the RAG search, bypassing expensive legal consultations.

For Managed Service Providers (MSPs) & Consulting Firms

- **Multi-Tenant Scale:** Enables a single consulting analyst to manage dozens of client accounts from one unified command dashboard.
- **Automated Client Deliverables:** Speeds up the creation of client-facing executive summaries through standard PDF reporting exports.
- **Client Acquisition and Scoping:** Uses interactive risk diagnostic surveys to evaluate a prospect's posture and immediately generate tailored mitigation playbooks.

5. Situational AI as a Strategic Multiplier

Traditional risk assessment is passive, relying on static compliance questionnaires. RiskPulse utilizes Situational AI to run active, dynamic threat modeling:

- **Adaptive Diagnostic Onboarding:** The AI Virtual CRO conducts an interactive diagnostic interview, analyzing answers to generate follow-up questions. It builds a SWOT analysis and three tailored playbooks specific to the business's size, model, and industry.
- **Live Threat Simulation Engine:** Users select scenarios from a library (e.g., Ransomware Extortion, Supply Chain Bankruptcy) to simulate a crisis. The engine maps the threat against the

company's actual cash, burn rate, MFA status, and CVE vulnerabilities to predict operational impacts and provide a mitigation roadmap.

- **Live Threat Intelligence Search:** Integrates live database search. If a new exploit or regulatory update goes live, the chatbot searches the web and interprets the relevance of that threat against the client's current balance sheet and IT assets.